

Introduction to Physics

Newton's Laws of Motion

First Law: An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.

Second Law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

Third Law: For every action, there is an equal and opposite reaction.

Key Concepts:

- Force: A push or pull upon an object
- Mass: The amount of matter in an object
- Acceleration: Rate of change of velocity